
朱雀
SUZAKU
CLOUD LOG DFIR TIMELINE GENERATOR
BLACK HAT USA 2026 · ARSENAL · YAMATO SECURITY

whoarewe?



Zach Mathis is the project leader for all the Yamato Security tools and resources. He has been doing security since the mid-90s and has been working professionally in Kobe, Japan since 2006. He has over three decades of experience with both red and blue team work and currently focuses on providing open source DFIR tools and resources. He has been teaching security in many countries throughout the APAC and EMEA regions since 2016.



Akira Nishikawa started his career as a software engineer specializing in embedded development. He worked as a freelance engineer in 2007, focusing on system development and operation for various companies. Since 2021, he has been dedicated to fostering a security culture for SaaS product security and improving service security. Additionally, he is an AWS Community Builder as of 2024.



Shota Nakajima (PINK) is a Tech Lead of Threat Intelligence at the Cyber Defense Institute. He specializes in malware analysis, with deep expertise in reverse engineering. In the field of threat intelligence, he actively tracks the latest APTs and has presented his extensive research at numerous international conferences, including JSAC, VB, HITCON, AVAR, CODE BLUE and Black Hat Arsenal.

Information Classification: General

Special Thanks!



Fukusuke Takahashi has been a member of NTTDATA-CERT (the CSIRT of NTT DATA Group Corporation) since 2018, specializing in IR, OSINT, and SOAR. He is also a member of Yamato Security and one of the core developers of its open-source tools. He has presented at several international conferences, including AUSCERT, SecTor, HITCON, the Annual FIRST Conference, SECCON and BlackHat.

Original Suzaku developer and has recently improved all the Sigma Azure rules so they can be used better natively!

Information Classification: General

What is Yamato Security?

- Started in 2012 to build a security community in Western Japan with free / low-cost workshops and training events
- Now one of the largest hands-on security communities in the country — 2500+ members (https://yamatosecurity.connpass.com)
- Releasing free OSS DFIR tools since 2021:
  - Hayabusa (Windows event log analyzer)
  - Takajo (Hayabusa results analyzer)
  - WELA (Windows Event Log Analyzer)
  - Yamato Security's Windows Event Log Configuration Guide For DFIR And Threat Hunting
  - Documentation: https://yamato-security.github.io/
  - Github: https://github.com/Yamato-Security

大和セキュリティ

Information Classification: General

New Tool Release!

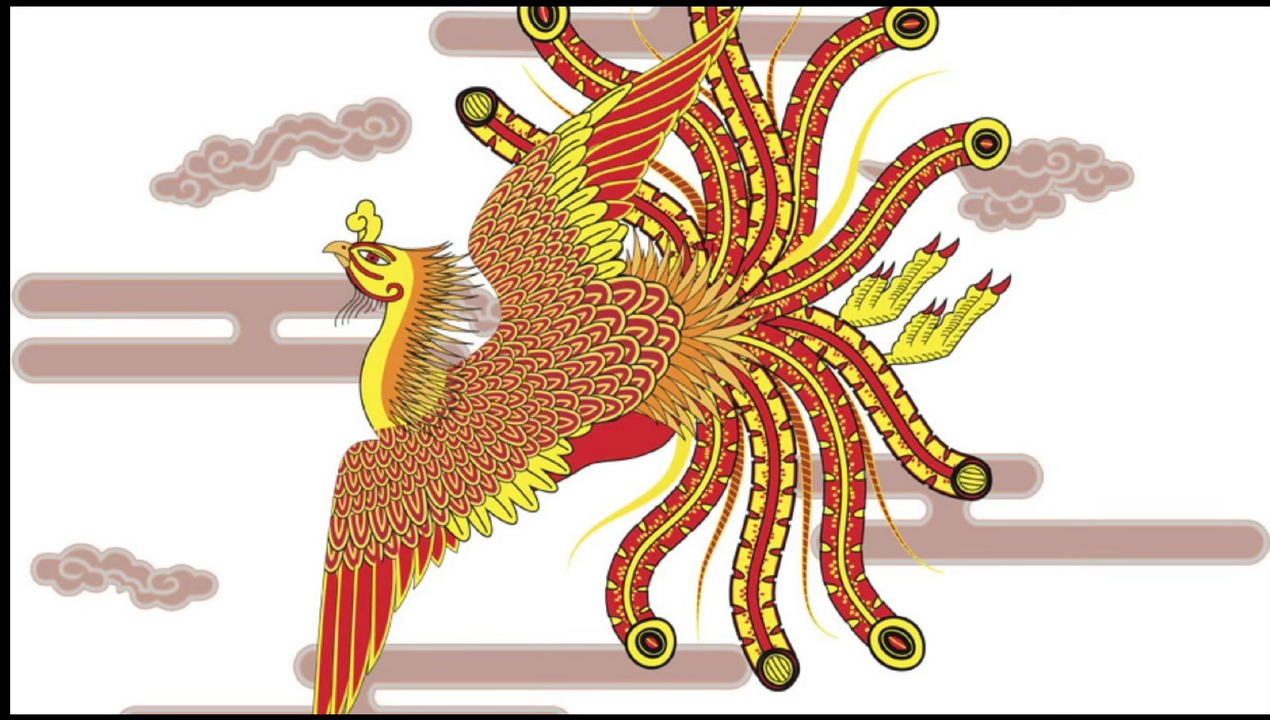
- Mecha Hayabusa official release tomorrow, also at Arsenal, 11:20am-12:20pm
  - AI automatic analysis of Hayabusa results
  - https://yamato-security.github.io/mecha-hayabusa/
  - https://github.com/Yamato-Security/mecha-hayabusa
- Senrigan (千里眼):
  - Offline, open-source AWS CloudTrail DFIR & threat hunting platform — 100+ built-in hunts, 80+ superset dashboard charts, AI chat, and an AWS Config resource graph.
  - https://yamato-security.github.io/senrigan/
  - https://github.com/Yamato-Security/senrigan
- Tool updates: Hayabusa BlackHat Release, etc…



Information Classification: General

Suzaku (朱雀)

- “Vermilion Bird”
- “Hayabusa” but for cloud logs
- Written in Rust
  (Fast, memory-safe, cross-platform, standalone binary)
- Native Sigma YML rules for detection
- Multi-thread support
- GeoIP location enrichment
- Supports AWS CloudTrail (json/json.gz) and now Azure logs as of v2.0
- Outputs to CSV/JSON(L) timelines
- License: AGPLv3



This release goes deeper on AWS and adds Azure for the first time.

Information Classification: General

Challenges with log analysis

- Many of the logs needed for a proper DFIR investigation are not enabled by default!
  - For cloud, management logs (ex: making an S3 bucket, etc…) are enabled by default but not data logs (ex: access to the S3 bucket)
  - Some organizations will only have logs enabled in certain regions
- Most of the logs are just noise and not relevant for DFIR
- There are no built-in logs that detect attacks
  - (Optional services like GuardDuty, MS Defender for Cloud, Google Cloud SCC, etc… exist but even those are not meant for thorough DFIR investigations)
- The output is too verbose and noisy
- The output is hard to parse (often needlessly complex JSON)
- Logs are separated into a large amount of multiple files
- No severity levels are set so it is hard to triage alerts

Information Classification: General

Solving these challenges

- Properly configuring your logs:
  - Prowler: auditing and properly enabling your cloud logs and much more!
    https://github.com/prowler-cloud/prowler
    https://prowler.com/
    OSS + paid tiers
  - AWS native: Security Hub's CIS AWS Foundations and Foundational Security Best Practices standards
- Solving the investigation challenges:
  customized output based on Sigma alerts, event summaries, etc…
  - Goal: Create a CSV/JSON timeline made for DFIR investigations that has only the important events and data, and can be easily triaged
  - ⁘Still a work in progress but constantly improving!

Information Classification: General

Sigma Rules

- The open, vendor-neutral standard for log-based detection.
- **Snort** for network · **YARA** for files · **Sigma** for logs
- Started late 2016. Now 500+ contributors and core team of maintainers
- Easy to read/write YML format
- Thousands of free, high-quality rules — often ahead of vendor detections
  - AttackRuleMap: https://attackrulemap.com
- No vendor lock-in: you can convert rules to any SIEM query (Splunk, Elastic, etc…)
- Detection Rule License (DRL) 1.1 — free to use, just credit the authors.
- https://github.com/sigmahq/sigma
- https://sigmahq.io/



Information Classification: General

Sigma Rule Example

```yaml
title: API Call From Hacking Distro
id: c85a99f2-bdfa-4d4a-8645-52628b2e4106
status: test
description: |
    An API call was invoked from a Linux hacking distribution (ex: Kali Linux, Parrot OS, etc.).
    When using the aws client, the default user agent is "aws-cli/{version} Python/{python_version} {platform}".
    This rule relies on the user agent string so it is easily bypassed by attackers who hardcode a benign-looing user agent in the botocore library.
references:
    - https://hackingthe.cloud/aws/general-knowledge/using_stolen_iam_credentials/
author: Zach Mathis (@yamatosecurity)
date: 2025-04-24
modified: 2025-04-24
tags:
    - attack.command-and-control
logsource:
    product: aws
    service: cloudtrail
detection:
    selection:
        userAgent|contains:
            - 'kali'
            - 'parrot'
            - 'pentoo'
    condition: selection
falsepositives:
level: medium
```

AWS Logs for DFIR

- Tier 1:
  - CloudTrail Management Events: Control Plane
  - CloudTrail Data Events: S3 bucket access, etc… Disabled.
  - GuardDuty Alerts
- Tier 2:
  - VPC Flow Logs: Detect lateral movement, exfil, etc… Disabled.
    v2 default format has no useful attribution fields so need to customize logs.
  - Route 53 Resolver Query Logs:
    Ex: detect C2 domain resolution, DNS tunneling, etc… Disabled.
  - EKS Control Plane Audit Logs: Kubernetes API audit delivered to CloudWatch logs.
    Needed for investigating container compromise. Disabled.

Information Classification: General

AWS Logs for DFIR

- Tier 3: (All disabled by default)
  - S3 Server Access Logs: Cheaper than normal S3 data access logs but best-effort, delayed and not guaranteed. Similar to space-delimited web access logs.
  - ELB/ALB Access Logs: Ex: Web app SSRF-to-IMDS compromise
  - WAF, Network Firewall Logs
  - CloudFront Access Logs: Needed when app is behind CDN and ALB logs only have CloudFront Ips
  - Lambda Execution Logs: CloudWatch (Somewhat enabled but useless without configuring)
  - SSM Session Management Logs: Ex: lateral movement via SSM
  - AWS Config Logs: Useful to know what the security settings were
  - Cost and Usage Report: Sometimes the best detection for cryptomining, etc…
  - ⁘Cost Explorer: By default, can infer attacks through spikes in cost

Information Classification: General

Azure Logs for DFIR

- Instead of one service (CloudTrail) that covers the entire control plane, Azure has at least four separate control planes!!!
(each with its own log, retention, licensing and export path…)

Plane	Log	Default Retention
Entra ID (identity)	SignInLogs, AuditLogs	7 days (30 with P1/P2)
Azure Resource Manager	Activity Log	90 days
M365	Unified Audit Log	180 days (varies by license)
Data Plane (per-resource)	Resource Logs	Disabled by default

Information Classification: General

Azure Logs for DFIR

Tier 1:

- Entra ID Sign-In Logs:
  - SignInLogs: Interactive user sign-ins
  - NonInteractiveUserSignInLogs: Token refresh, client apps
    Usually 10-50x volume but useful for finding token-theft and stolen-refresh-token activity
  - ServicePrincipal: App / SP authentication.
    Can find OAuth abuse and stolen client secrets, etc…
  - ManagedIdentitySignInLogs:
    Detecting managed identity abuse. SSRF+IMDS exploits, etc…

Information Classification: General

Azure Logs for DFIR

Tier 1:

- Entra ID Audit Log: Detect persistence mechanisms -
  adding service principals, consenting to apps, adding members to roles, MFA changes, updating conditional access policies, etc…
- Azure Activity Log: ARM control plane – VM creation, role assignments, key vault access policy changes, disk snapshot/export, NSG changes, etc…
- M365 UAL (Unified Audit Log):
  Mailbox rules/items accessed, SharePoint file access, etc…

Information Classification: General

Azure Logs for DFIR

Tier 2:

- Key Vault Resource Logs: Needed for secrets exfil evidence. Disabled.
- Storage Account Resource Logs: Similar to S3 data events. Disabled.
- NSG Flow Logs / VNet Flow Logs: Lateral movement, exfil, etc…
- AKS Control Plane Audit: Needed for container DFIR. Disabled.
- MS Defender for Cloud Alerts: Similar to GuardDuty. Not free.
- MS Graph Activity Logs: Disabled by default. Important for detecting enumeration, dumping email via Graph with stolen tokens, etc…
- Provisioning Logs: Entra-to-SaaS sync abuse, etc…
- Identity Protection: RiskyUsers / UserRiskEvents / RiskDetections

Information Classification: General

Azure Logs for DFIR

Tier 3:

- Azure Firewall
- DNS Proxy Logs
- Application Gateway
- WAF Logs
- Front Door Logs
- Azure SQL Audit Logs
- App Service HTTP Logs
- Cost Management Logs (Ex: detecting cryptominers, etc…)

Information Classification: General

Downloading the logs (AWS)

- Make sure you are forwarding your CloudTrail logs to a private S3 bucket or else you will only have 90 days of logs and a very difficult time downloading them!
- Downloading from S3:
  - `aws s3 cp s3://<your-bucket-prefix> <local-output-dir>/ --recursive --exclude "*" --include "*.json.gz"`
  - Invictus-AWS: Acquire data from all regions and other triage ability https://github.com/invictus-ir/Invictus-AWS
  - CloudTrail Log Explorer https://github.com/dlcowen/CloudTrailLogExplorer
- Downloading in chunks via CloudTrail API from default location:
  - AWS Cloud Trail Downloader V2: https://www.hecfblog.com/2024/09/aws-cloud-trail-downloader-v2.html
    Python script: https://raw.githubusercontent.com/dlcowen/sansfor509/main/AWS/Cloudtrail_downloadv2.py

Information Classification: General

Downloading the logs (Azure)

- Azure log formats and field names change depending on the export method!!!
- Main methods:
  1. API pull — JSON. Graph (Entra) or ARM (Activity Log): different APIs, different schemas. Most IR engagements will have to analyze these logs as they are on by default.
  2. Diagnostic settings export (Same schema, different encoding)
     - => Storage Account => JSONL (Many PT1H.json blob files)
     - => Event Hub => JSON records (Streaming to a non-Sentinel SIEM)
     - Sigma now standardizes on this format.
  3. Log Analytics / KQL export — CSV/JSON. Sentinel tenants. PascalCase.
  4. Azure Portal CSV — Lossy. Avoid.
- Retention is the real constraint — 7 days Entra / 90 days Activity Log unless export was configured in advance.

Information Classification: General

Downloading the logs (Azure)

- API Pulling Tools:
  - Microsoft-Extractor-Suite (invictus-ir): Sign-In, Audit, Activity, UAL, Risky, etc… logs and outputs to JSON(L)/CSV/SOF-ELK. Maintained well.
    - PowerShell
    - https://github.com/invictus-ir/Microsoft-Extractor-Suite
  - Untitled Goose Tool (CISA + Sandia):
    - Python
    - https://github.com/cisagov/untitledgoosetool
- Downloading PT1H.json files from blob storage:
  - `az storage container list --account-name <acct> --auth-mode login -o table`
  - `azcopy copy "https://<acct>.blob.core.windows.net/insights-logs-signinlogs?<SAS>" "./azure-logs/" --recursive`

Information Classification: General

Sigma Rules for Cloud Logs

- Rules hosted at: https://github.com/Yamato-Security/suzaku-rules
- Native support so no conversion needed (nor ”loss in translation”)!
- Support for v2 specifications
- Correlation rules to detect more advanced attacks across multiple logs and reduce false positives
- Official rules are synced to the suzaku-rules repository every day and you can easily sync them locally with the `update-rules` command

Information Classification: General

Sigma Rules for Cloud Logs

888

Sigma rule files (as of July 2026 — synced daily)

BY SOURCE

742 Suzaku built-in (83.6%)

146 Upstream Sigma (16.4%)

BY CLOUD

791 AWS (89.1%)

76 Azure (8.6%)

21 M365 (2.4%)

Information Classification: General

Category	Rules	%	Upstream	Suzaku
aws / cloudtrail	791	89%	53	738
azure / activity_logs	37	4%	33	4
azure / identity_protection	19	2%	19	0
m365 / threat_management	13	1.5%	13	0
azure / audit_logs	8	1%	8	0
azure / privileged_identity_management	7	1%	7	0
m365 / audit	6	1%	6	0
azure / signin_logs	5	0.5%	5	0
m365 / exchange	1	0.1%	1	0
m365 / threat_detection	1	0.1%	1	0

Information Classification: General

01 - Leaks / Compromised keys

How it works

- Keys leak via git, .env files, CI logs, screenshots
- Attacker confirms access with GetCallerIdentity
- Pivots fast from an unfamiliar IP and geo

In the Suzaku timeline

- API calls from a new country / Kali / Pacu UA
- AWS attaches AWSCompromisedKeyQuarantine on leak

Rules

```
API Call From Hacking Distro
Access Key Quarantined by AWS
```

02 - Recon & Enumeration

How it works

- Maps identities, roles, buckets, secrets
- IAM / STS / S3 List* and Get* calls in bursts
- Trips AccessDenied while probing permissions

In the Suzaku timeline

- Dozens of enumeration calls from one key
- A spike of AccessDenied from one principal

Rules

```
Many Recon Events
Many AccessDenied From Single Source
```

03 - Privilege Escalation

How it works

- Attaches AdministratorAccess to itself
- Creates a new default policy version / inline policy
- Abuses iam:PassRole into a privileged role

In the Suzaku timeline

- Attach*Policy with an admin policy ARN
- CreatePolicyVersion / SetDefaultPolicyVersion

Rules

`IAM Admin Policy Attached`

`IAM Privilege Escalation Primitive`

04 - SSRF to Instance metadata

How it works

- App-side SSRF reaches 169.254.169.254
- Steals the EC2 instance-role credentials
- Replays them off-box (the Capital One pattern)

In the Suzaku timeline

- Instance-role session (i-...) from a non-VPC IP
- GeoIP shows a source outside your network

Rules

```
Instance Creds Used Outside VPC
```

05 - Persistence / Backdoors

How it works

- Creates a new IAM user and access key
- Adds a console login profile to a victim user
- Adds cross-account trust to an attacker account

In the Suzaku timeline

- CreateUser then CreateAccessKey, same source
- UpdateAssumeRolePolicy adds a foreign account

Rules

`IAM Persistence Chain (temporal)`

`Role Trust Policy Modified`

06 - Defense Evasion

How it works

- Stops or deletes the CloudTrail trail
- Disables the GuardDuty detector
- Deletes VPC flow logs / Config recorder

In the Suzaku timeline

- cloudtrail StopLogging / DeleteTrail
- guardduty DeleteDetector / disable

Rules

`CloudTrail Logging Disabled`

`GuardDuty Detector Disabled`

07 - S3 Data Exfiltration

How it works

- Makes a bucket or object public, or
- Bulk-downloads objects to attacker storage

In the Suzaku timeline

- DeletePublicAccessBlock / public-read ACL
- Thousands of GetObject from one key

Rules

`S3 Bucket Made Public`

`S3 Mass Download (event_count)`

08 - Snapshot & AMI Sharing

How it works

- Shares an EBS / RDS snapshot or AMI
- Targets an external account or ‘all’
- Restores it on their side to read the data

In the Suzaku timeline

- ModifySnapshotAttribute adds group ‘all’
- or an unknown destination account id

Rules

```
Snapshot or AMI Shared
Externally
```

09 - Cryptomining / Hijacking

How it works

- Launches GPU fleets with stolen creds
- Spreads across regions to avoid notice
- Instances beacon out to mining pools

In the Suzaku timeline

- RunInstances of p* / g* GPU types
- Many launches fast; GuardDuty CryptoCurrency

Rules

```
GPU Instance Launched

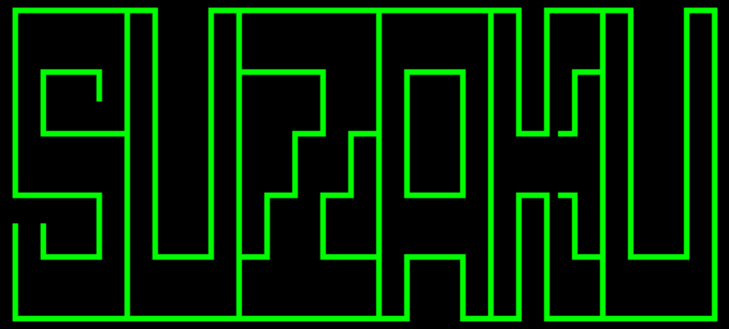
Mass RunInstances
(event_count)
```

Commands

- **aws-ct-timeline:** Creates an AWS CloudTrail DFIR timeline
- **aws-ct-search:** Search AWS CloudTrail logs
- **aws-ct-metrics:** Generates metrics from AWS CloudTrail logs
- **aws-ct-summary:** Generates summary from AWS CloudTrail logs
- **azure-timeline:** Creates an Azure DFIR timeline
- **update-rules:** Update rules
- Output to CSV and/or JSONL for easy viewing or importing to a SIEM

Information Classification: General

DEMO TIME



by Yamato Security

```
Start time: 2025/04/19 09:07

Total detection rules: 46
Total log files: 3975
Total file size: 251.6 MiB

[00:02:23] 3,975 / 3,975   [=======================================]

Scanning finished.
```

Information Classification: General

2024-08-02 08:53:24 · Console Login Without MFA · Zach Mathis (@yamatosecurity) · medium · ConsoleLogin · signin.amazonaws.com · eu-west-2r · 225.01.00.16 · stratus-red-team_fccf7123-0651-41f5-b06c-460da5ee1c94 · stratus-red-team-nmfalu-jfzdtsvchl · IAMUser · 562283505220 · arn:aws:iam::562283505220:user/stratus-red-team-nmfalu-jfzdtsvchl · AIDA1ERT0661IN5R239V · - · 865d9377-9c6b-4fd7-8aad-725e95f6a140 · 4ae38c23-b180-4120-87bf-7bf5cb18dd5b

2024-08-01 11:30:23 · AWS STS AssumeRole Misuse · Austin Songer @austinsonger · low · RunInstances · ec2.amazonaws.com · ca-south-3r · 06.237.252.245 · stratus-red-team_c8ff220a-7e52-429b-868f-d979123ed2d3 · - · AssumedRole · 900138736586 · arn:aws:sts::900138736586:assumed-role/stratus-red-team-ec2lui-role-idtzskbvtd/aws-go-sdk-1722511821294449000 · AROA13YEHY3VAS32TD341:aws-go-sdk-1722511821294449000 · ASIA9F6MXE9HSYOXYQOS · 1a4debbb-12e9-4bde-b8c7-ea29002bb2a7 · 905d389b-b853-46d0-9d3d-dea0d3a3cd49

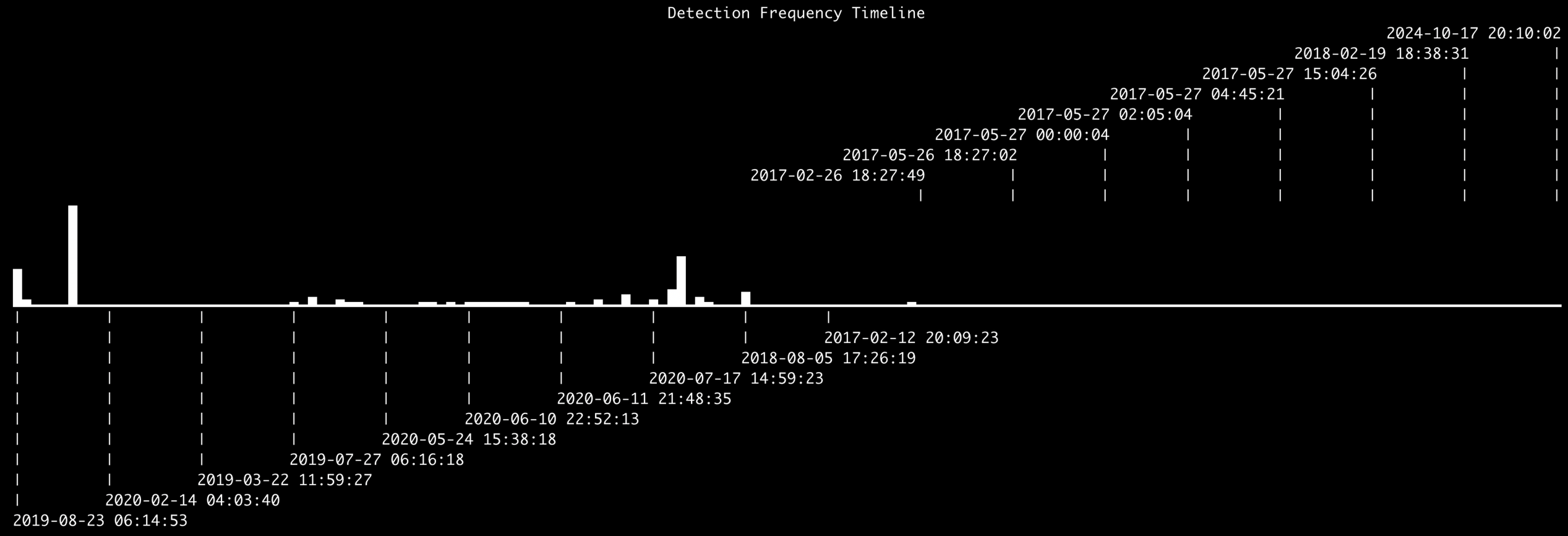
2024-07-31 12:46:41 · CloudTrail Log Deleted · Zach Mathis (@yamatosecurity) · medium · DeleteTrail · cloudtrail.amazonaws.com · megov-westwest-1r · 08.1.250.216 · stratus-red-team_a007fa03-86e2-4130-be03-ee7b7b10edcc · christophe · IAMUser · 847129010505 · arn:aws:iam::847129010505:user/christophe · AIDALE4EP1EPEPX3SDR8 · AKIAFBJ48BV9CGRBRKGM · ee73c230-44bc-4492-8542-cfb189eae287 · fa5c25ee-9c78-4af4-916f-52a81d1d16e6

2024-07-31 13:06:24 · CloudTrail Logging Stopped · Zach Mathis (@yamatosecurity) · high · StopLogging · cloudtrail.amazonaws.com · apiso-centralnorth-2r · 86.245.153.234 · stratus-red-team_c97089f1-1ae3-4ecc-b006-f5e8fd0f2571 · christophe · IAMUser · 143434273843 · arn:aws:iam::143434273843:user/christophe · AIDAOC1SYDVN0AF0FMMR · AKIAGGWFBBHBE7D3M9WI · 10163ed2-2253-469d-a5ee-cbc6651f8934 · da90c11e-e3d4-411e-aaf7-06d02e0c5c23

2024-08-02 08:30:00 · AWS STS AssumeRole Misuse · Austin Songer @austinsonger · low · LeaveOrganization · organizations.amazonaws.com · euiso-south-3r · 252.5.222.230 · stratus-red-team_fd969928-3c0d-4feb-bd56-34f9aee3e6eb · - · AssumedRole · 307578594326 · arn:aws:sts::307578594326:assumed-role/stratus-red-team-leave-org-role/aws-go-sdk-1722587398902687000 · AROAHKPEEQ9BHUOX4D93T:aws-go-sdk-1722587398902687000 · ASIA36EV31F1RB3OA8IG · 16903cbd-fdff-4818-82f2-d66ad09aaf57 · 905d389b-b853-46d0-9d3d-dea0d3a3cd49

2024-08-01 12:23:55 · Ingress/Egress Security Group Modification · jamesc-grafana · medium · AuthorizeSecurityGroupIngress · ec2.amazonaws.com · us-northeast-1r · 253.243.215.253 · stratus-red-team_1004a4ff-b486-4981-a84b-6322905f37cc · christophe · IAMUser · 032092706103 · arn:aws:iam::032092706103:user/christophe · AIDAQ5Y2TGCDATQV6SRP · AKIAXW7UJ577KFYIAHIM · 9fd68588-ecbf-4528-a345-199fa6bb0821 · 6fb77778-040f-4015-9440-572aa9b6b580

Information Classification: General



Information Classification: General

Results Summary:
Events with hits / Total events: 66,690 / 1,972,854 (Data reduction: 1,906,164 events (96.62%))
Total | Unique critical detections: 0 (0%) | 0 (0%)
Total | Unique high detections: 2,577 (3%) | 3 (0%)
Total | Unique medium detections: 11,933 (17%) | 9 (0%)
Total | Unique low detections: 55,017 (82%) | 4 (0%)
Total | Unique informational detections: 63 (0%) | 1 (0%)

First event time: 2017-02-12 19:57:06 UTC
Last event time: 2024-10-17 20:11:24 UTC

Dates with most total detections:
critical: n/a, high: 2020-10-06 (1,102), medium: 2017-02-19 (1,673), low: 2017-05-27 (12,977), informational: 2017-09-17 (2)

Top critical alerts:	Top high alerts:
n/a	Malicious Usage Of IMDS Credentials Outside Of AWS Infrastructure (2,573)
n/a	AWS User Login Profile Was Modified (3)
n/a	AWS Config Disabling Channel/Recorder (1)
n/a	n/a
n/a	n/a

Top medium alerts:	Top low alerts:
AWS Root Credentials (11,722)	AWS STS AssumeRole Misuse (42,472)
AWS Snapshot Backup Exfiltration (59)	Potential Bucket Enumeration on AWS (12,134)
AWS IAM Backdoor Users Keys (56)	AWS STS GetSessionToken Misuse (209)
Console Login Without MFA (26)	AWS S3 Data Management Tampering (202)
AWS CloudTrail Important Change (23)	n/a

Top informational alerts:
Console Login With MFA (63)
n/a
n/a
n/a
n/a

Elapsed time: 00:02:41

Information Classification: Gene

black hat USA 2026

Testing Suzaku

- https://github.com/Yamato-Security/suzaku-sample-data
  - Over 4000+ sample logs!
  - Invictus-ir dataset: In August 2023, Invictus-ir did great research on simulating 32 attacks against AWS with the Stratus Red Team tool, comparing Sigma detections (run via Athena and Splunk) against GuardDuty, and shared the dataset after the attacks.
  - Flaws.cloud dataset: Scott Piper from Summit Route made his public data from attacks against his free flaws.cloud training website in 2020.
  - SANS 504 S3 Ransomware Lab: This dataset comes from a simulated S3 ransomware scenario that was used in a previous lab. This is released with permission from course author Josh Wright.
  - Stratus Red Team: A great project by Datadog that simulates cloud attacks. Logs are automatically generated by Grimoire and anonymized by LogLicker.
  - Traildiscover.cloud: A great website that lists up all of the AWS API calls used in attacks and provides sample JSON data.

Information Classification: General

Contributing

- We welcome anyone to help out with our projects!
- In particular:
  - Simulating cloud attacks and providing sample logs
  - Writing Sigma detection rules
  - Writing documentation for configuring and exporting logs
  - Help with mapping and converting the different Azure log formats and fields
  - Behavioral analysis to complement signature detection
  - Write-ups on importing Suzaku output into SIEMs
  - Help with GCP support
  - Testing, feedback and new ideas! Create an issue or discussion on GitHub!
  - Give us a GitHub star and/or post on X / Twitter!

Information Classification: General



(千里眼 = ”The 1000 mile-seeing eye” => “Clairvoyance”) —
Yamato Security's offline, open-source AWS CloudTrail DFIR & threat-hunting platform

Information Classification: General

Senrigan

- Get it here: https://github.com/Yamato-Security/senrigan
- What it is:
  - Hunt AWS threats in minutes — no SIEM, no cloud infrastructure required
  - Runs entirely offline on a single laptop
  - Air-gapped with a single `make up` (Docker Compose)
  - Open source (AGPL-3.0), released by Yamato Security in 2026 at DEF CON!
  - Docs in 15 languages
- Example hunts / detections:
  - Off-hours writes, reconnaissance bursts, multi-region spread, first-time API calls
  - IAM enumeration & credential access, unusual/attacker-tool user agents (Pacu, curl, etc…)
  - Temporal analytics: first/last-seen, dormant accounts reactivated, velocity spikes

Information Classification: General

Senrigan

- What's inside:
  - Drop in CloudTrail logs → Start hunting!
  - 100+ built-in threat hunts — ready-to-run queries; pick one from a dropdown, no SQL or deep AWS knowledge needed
  - 80+ pre-built Apache Superset dashboard charts — BI visualization of API activity, identities, source IPs, user agents, temporal patterns
  - AI-assisted chat — natural-language threat hunting that generates the queries for you (optional: OpenAI key)
  - AWS Config resource graph — visualize your AWS resources/relationships from Config snapshots
  - HTML threat-hunting reports + Suzaku results visualization

Information Classification: General

Senrigan

- Architecture:
  - A fast Rust ingester is the only writer to the backend DuckDB database. Every analysis component (dashboard, AI agent) queries it read-only to prevent accidentally overwriting any data.
  - Modular Docker stack: ingester (Rust) · agent (Python / Streamlit AI chat) · dashboard (Apache Superset) · config_viz (resource graph)
  - http://localhost:8501 — Built-in queries and AI Chat
  - http://localhost:8088 — Dashboard (admin / admin)
  - http://localhost:8502 — AWS Config resource graph

Information Classification: General

DEMO TIME

Information Classification: General

Senrigan

localhost:8501/suzaku_ct_summary

シークレッ

Senrigan

Suzaku CT Summary

Summary File

aws-ct-summary JSON

sample...ummary.json
4.2MB

Report

Markdown | HTML

Session

Export JSON | Clear

Inspect identity

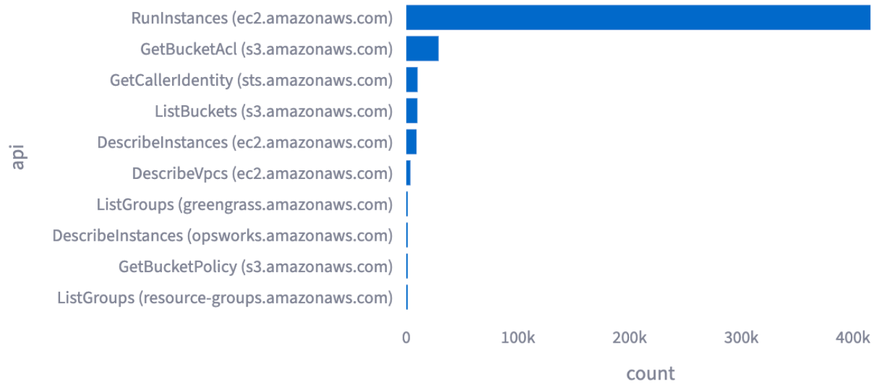
arn:aws:iam::123456789012:user/backup — IAMUser · 680,449 events · abused 11✅/23❌

arn:aws:iam::123456789012:user/backup

Type	Total events	First seen	Last seen
IAMUser	680,449	2017-02-12 21:15:12	2020-10-07 21:03:30

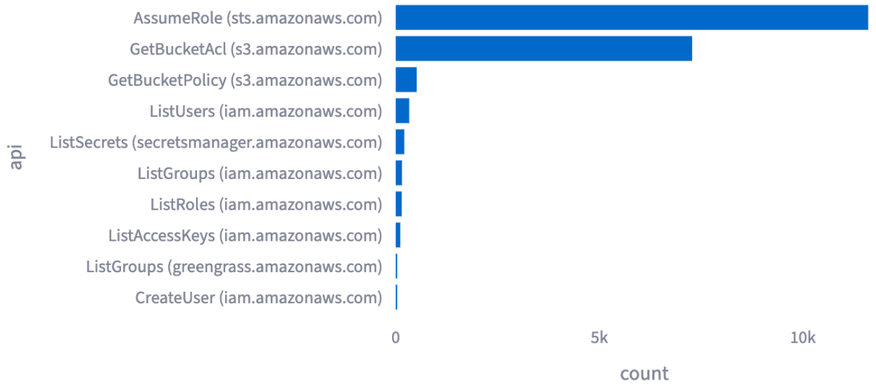
Abused APIs

✅ Succeeded



api	description	count	first_seen	last_seen
RunInstances (ec2.amazonaws.com	Spin up EC2 instances (c	415552	2019-08-23 06:00:07	2019-08-2
GetBucketAcl (s3.amazonaws.com)	S3 recon	28907	2019-08-21 08:03:03	2019-10-2
GetCallerIdentity (sts.amazonaws.(	Current credentials reco	10095	2019-08-23 06:00:07	2019-08-2
ListBuckets (s3.amazonaws.com)	S3 recon	9936	2019-08-23 06:00:07	2019-08-2
DescribeInstances (ec2.amazonaw	EC2 and network layout	9168	2019-08-23 06:00:07	2019-08-2
DescribeVpcs (ec2.amazonaws.com	EC2 and network layout	3658	2019-08-21 08:03:03	2019-09-1

❌ Failed



api	description	count	first_seen	last_seen
AssumeRole (sts.amazonaws.c	Lateral movement via rol	11603	2019-08-21 08:03:03	2019-09-18 07
GetBucketAcl (s3.amazonaws.c	S3 recon	7279	2019-08-21 08:03:03	2019-09-09 09
GetBucketPolicy (s3.amazonaw	S3 recon	515	2019-08-21 08:03:03	2019-10-01 19
ListUsers (iam.amazonaws.com	IAM enumeration	331	2019-08-21 08:03:03	2019-08-29 14
ListSecrets (secretsmanager.an	Find credential storage lo	210	2019-08-21 08:03:03	2019-10-19 23
ListGroups (iam.amazonaws.co	IAM enumeration	153	2019-08-21 08:03:03	2019-09-12 15

Information

hat

Config Viz — AWS Resource

localhost:8502

シークレット

Config Viz

AWS Resource Graph

Search resources...

Layout Direction

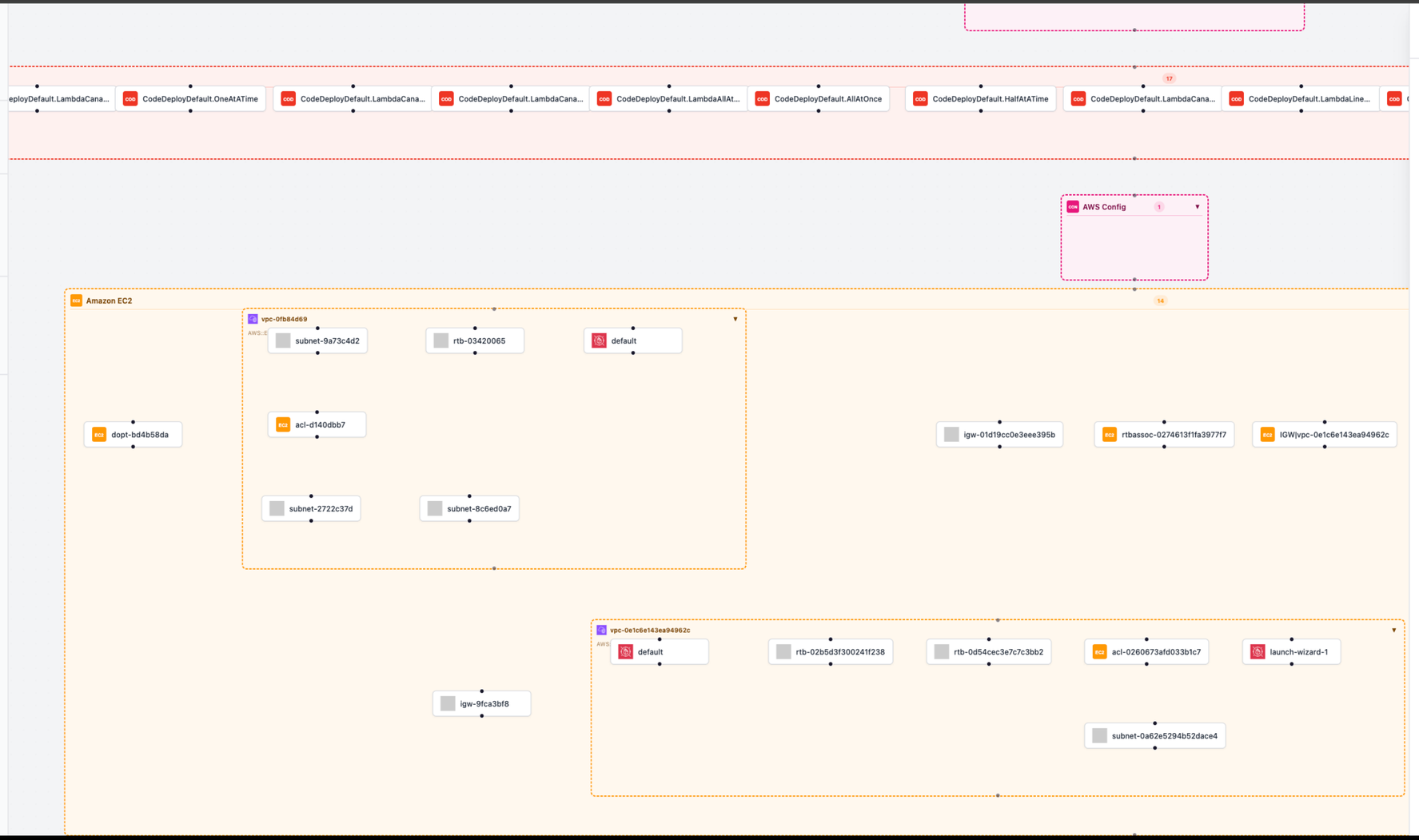
TB | LR

Filter by Type

All types

Snapshots

4abcbf08-4589-4760-b9ca-45e...
ap-northeast-1 · 134 resources



vpc-0e1c6e143ea94962c

Info

Type	AWS::EC2::VPC
Region	ap-northeast-1
ID	vpc-0e1c6e143ea94962c

Tags

Name	hayabusa-vpc

Configuration


```
{
  "cidrBlock": "192.168.0.0/24",
  "cidrBlockAssociationSet": [
    {
      "associationId": "vpc-cidr-assoc-0fe2
dd1b632b3e11",
      "cidrBlock": "192.168.0.0/24",
      "cidrBlockState": {
        "state": "associated"
      }
    }
  ],
  "dhcpOptionsId": "dopt-bd4b58da",
  "instanceTenancy": "default",
  "ipv6CidrBlockAssociationSet": [],
  "isDefault": false,
  "ownerId": "015043439996",
  "state": "available",
  "tags": [
    {
      "key": "Name",
      "value": "hayabusa-vpc"
    }
```

CloudTrail Threat Hunting

localhost:8088/superset/dashboard/cloudtrail-threat-hunting/?native_filters_key=wNjjs9CfjIs

シークレット

Superset | Dashboards | Charts | Datasets | SQL | Settings

Filters and controls

Date Range: No filter

Identity Type: 6 options

Identity Type NOT: is not | 6 options

Principal ARN + Include: 58 options

Principal ARN NOT: is not | 58 options

Source IP + Include: 1000 options

Source IP NOT: is not | 1000 options

Event Name + Include: 1000 options

Event Name NOT: is not | 1000 options

Event Source + Include: 164 options

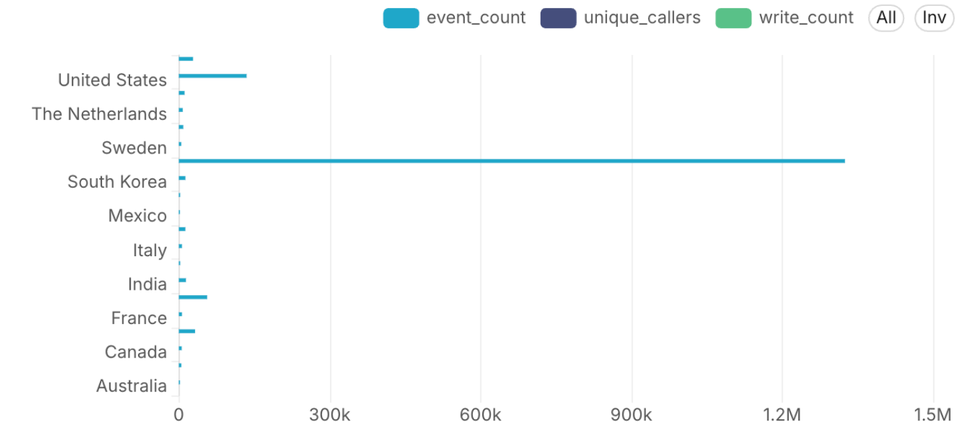
Apply filters

CloudTrail Threat Hunting

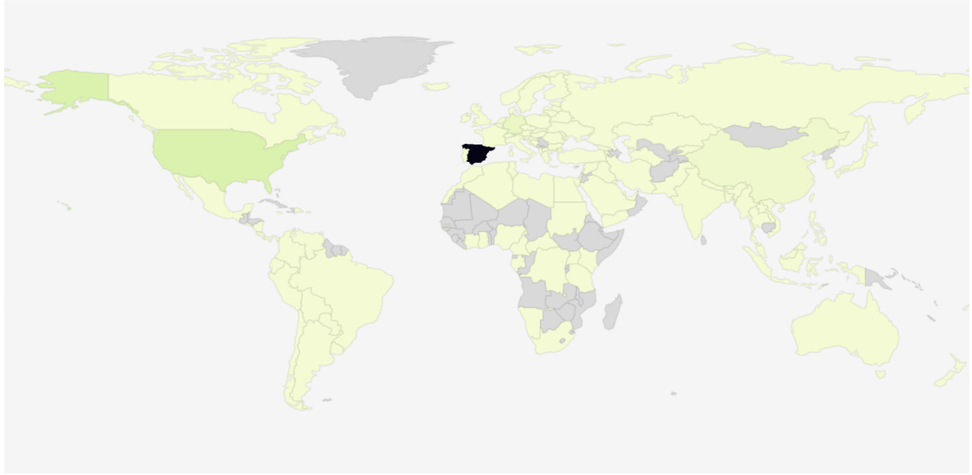
Published | Admin User | 22 minutes ago | Edit dashboard

Identity & Access | Threat Detection | API Activity | Computing | S3 & RDS | GeoIP Intelligence | Temporal Analysis | High-Risk API Monitor

Top Countries by Request Volume



Global Request Origin Map



Top Cities by Request Volume

geo_city	geo_country_name	request_count	write_requests	unique_identities
Madrid	Spain	1.32M	0	4
Scottsdale	United States	10.2k	0	2
Elizabethton	United States	7.11k	0	1
Daan	Taiwan	7.08k	0	1
Portland	United Kingdom	5.14k	0	1
Toronto	Canada	4.59k	0	5
Rome	Italy	4.58k	0	4
Ashburn	United States	4.3k	0	5

Top ASN Organizations by Request Volume

geo_org	request_count	write_requests	unique_identities
Telefonica De Espana S.a.u.	1.32M	0	4
GRASS-MERKUR GmbH & Co. KG	47k	0	1
United States Department of Defense DoD	24.2k	9	6
CHINA UNICOM China169 Backbone	18.4k	0	5
Amazon.com, Inc.	17.9k	2	6
Cox Communications Inc.	11k	0	6
Microsoft Corporation	10.1k	1	6
Taiwan Academic Network TANet Information Center	8.04k	0	2

Expand all Collapse all
🔑 Identity & Access
🛡️ Detection & Response
🪣 Data & Storage
🌐 Network & Infrastructure
🥷 Threat Patterns
🌙 Off-Hours Activity (200 rows)
🔍 Reconnaissance Pattern (200 rows)
🌍 Multi-Region Activity (0 rows)
🤖 Unusual User Agents (200 rows)
📊 Activity & Baseline
⚡ Compute & Serverless
☁️ IaC & Platform
🌍 GeoIP Analysis

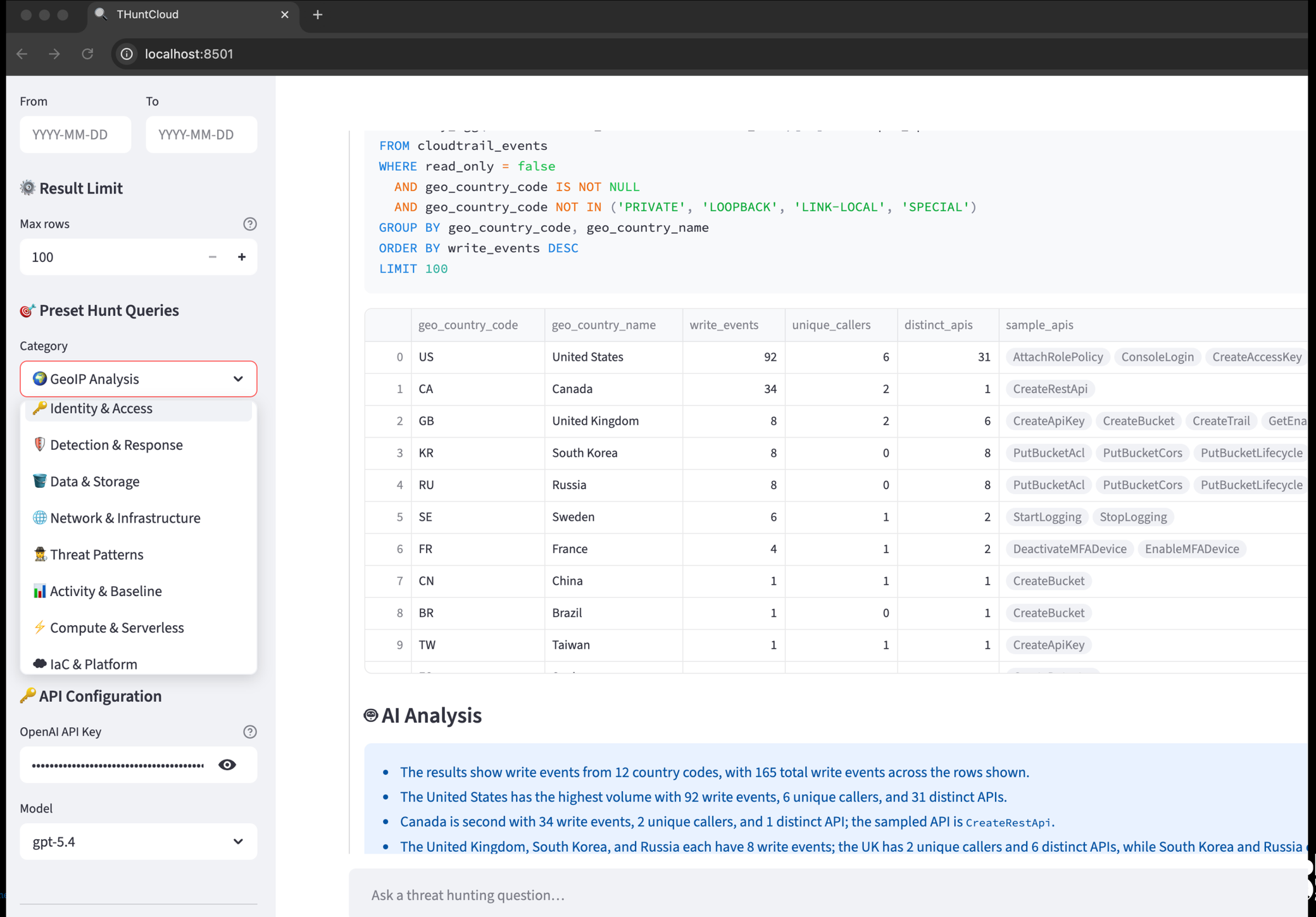
Query 43 — 🥷 Threat Patterns › 🤖 Unusual User Agents (200 rows)

SQL

```
SELECT
    user_agent,
    COUNT(*)                        AS event_count,
    COUNT(DISTINCT user_identity_arn) AS distinct_callers,
    MIN(event_time)                 AS first_seen,
    MAX(event_time)                 AS last_seen
FROM cloudtrail_events
WHERE user_agent IS NOT NULL
GROUP BY user_agent
HAVING COUNT(*) < 5
    OR lower(user_agent) LIKE '%pacu%'
    OR lower(user_agent) LIKE '%boto%'
    OR lower(user_agent) LIKE '%python%'
    OR lower(user_agent) LIKE '%curl%'
    OR lower(user_agent) LIKE '%wget%'
ORDER BY event_count ASC
LIMIT 200
```

Results

user_agent	event_count	distinct_callers	first_seen	last_seen
[aws-cli/2.0.39 Python/3.8.5 Darwin/19.5.0 source/x86_64 command/s3.ls]	1	1	2020-08-12 09:50:00	2020-08-12 09:50:00
[aws-cli/1.16.290 Python/3.7.5 Linux/5.3.12-200.fc30.x86_64 botocore/1.13.26]	1	1	2019-11-28 23:18:18	2019-11-28 23:18:18
[aws-cli/1.16.196 Python/3.6.0 Windows/10 botocore/1.12.186]	1	1	2019-11-22 06:10:52	2019-11-22 06:10:52
[aws-cli/1.16.92 Python/3.6.8 Linux/4.19.0-kali1-686-pae botocore/1.12.82]	1	1	2019-01-23 22:36:32	2019-01-23 22:36:32
[aws-cli/1.16.160 Python/3.7.3 Darwin/18.7.0 botocore/1.12.150]	1	1	2019-12-12 20:41:18	2019-12-12 20:41:18
[aws-cli/1.18.2 Python/3.7.6 Darwin/19.3.0 botocore/1.15.2]	1	1	2020-02-21 01:25:32	2020-02-21 01:25:32
[aws-cli/2.0.5 Python/3.7.4 Darwin/19.4.0 botocore/2.0.0dev9]	1	1	2020-04-15 22:50:02	2020-04-15 22:50:02
[aws-cli/2.0.1 Python/3.7.4 Darwin/19.3.0 botocore/2.0.0dev5]	1	1	2020-03-11 12:23:15	2020-03-11 12:23:15


THuntCloud
localhost:8501
From
To
YYYY-MM-DD
YYYY-MM-DD
Result Limit
Max rows
100
Preset Hunt Queries
Category
GeoIP Analysis
Identity & Access
Detection & Response
Data & Storage
Network & Infrastructure
Threat Patterns
Activity & Baseline
Compute & Serverless
IaC & Platform
API Configuration
OpenAI API Key
Model
gpt-5.4
FROM cloudtrail_events
WHERE read_only = false
AND geo_country_code IS NOT NULL
AND geo_country_code NOT IN ('PRIVATE', 'LOOPBACK', 'LINK-LOCAL', 'SPECIAL')
GROUP BY geo_country_code, geo_country_name
ORDER BY write_events DESC
LIMIT 100
geo_country_code | geo_country_name | write_events | unique_callers | distinct_apis | sample_apis
0 | US | United States | 92 | 6 | 31 | AttachRolePolicy ConsoleLogin CreateAccessKey
1 | CA | Canada | 34 | 2 | 1 | CreateRestApi
2 | GB | United Kingdom | 8 | 2 | 6 | CreateApiKey CreateBucket CreateTrail GetEna
3 | KR | South Korea | 8 | 0 | 8 | PutBucketAcl PutBucketCors PutBucketLifecycle
4 | RU | Russia | 8 | 0 | 8 | PutBucketAcl PutBucketCors PutBucketLifecycle
5 | SE | Sweden | 6 | 1 | 2 | StartLogging StopLogging
6 | FR | France | 4 | 1 | 2 | DeactivateMFADevice EnableMFADevice
7 | CN | China | 1 | 1 | 1 | CreateBucket
8 | BR | Brazil | 1 | 0 | 1 | CreateBucket
9 | TW | Taiwan | 1 | 1 | 1 | CreateApiKey
AI Analysis
The results show write events from 12 country codes, with 165 total write events across the rows shown.
The United States has the highest volume with 92 write events, 6 unique callers, and 31 distinct APIs.
Canada is second with 34 write events, 2 unique callers, and 1 distinct API; the sampled API is CreateRestApi.
The United Kingdom, South Korea, and Russia each have 8 write events; the UK has 2 unique callers and 6 distinct APIs, while South Korea and Russia
Ask a threat hunting question…

Future Plans

- Support other AWS logs besides CloudTrail
- GCP and other cloud log support
- Triage and detection assisted by GenAI
- Support for Azure logs in Senrigan
- Tool for checking and downloading all the Azure logs…

Information Classification: General

Summary

- Suzaku is now good at: AWS DFIR
- Also check out for AWS DFIR: Senrigan
- Work in progress: Azure DFIR
- This is an open-source community project so feel free to help the community!
- Feel free to open an issue / PR / discussion on GitHub!
- Thanks for coming to listen! ご静聴、ありがとうございました！

Information Classification: General